Determining Default Probabilities from CDS Premiums

The Assignment

Your analysts obtain the following information:

- The set of CDS premiums for XYZ plc for each six-month period for the next five years
- The set of six-month forward rates

(You may assume that each six-month period is exactly 0.5 year long.)

Your analysts also estimate that the recovery rate is 40%.

Period	Forward rates	CDS Premiums
0.5	2.25%	8.00%
1.0	2.50%	7.50%
1.5	2.88%	7.00%
2.0	3.00%	6.80%
2.5	3.25%	6.50%
3.0	3.50%	6.00%
3.5	4.00%	5.50%
4.0	4.38%	5.25%
4.5	4.75%	5.00%
5.0	5.00%	4.80%

REQUIRED

Using the spreadsheet template supplied, calculate:

1. The set of discount factors at the end of each six-month period.
2. The set of default probabilities for each six-month period.
3. The set of cumulative survival probabilities at the end of each six-month period.

USEFUL FORMULA

The following formula allows you to bootstrap the default probabilities from the swap premiums:

$$PD_N = \frac{s_N\left[\sum_{j=1}^{N-1} v_j T_{j-1,j} PS_j + v_N T_{N-1,N} PS_{N-1}\right] - (1-R)\sum_{j=1}^{N-1} v_j PD_j}{v_N\left[s_N T_{N-1,N} + (1-R)\right]}$$

Where:

s_j	is the swap premium for period j
v_j	is the discount factor for period j
PD_j	is the probability of default in period j
PS_j	is the probability of survival through period j
$T_{j-1,j}$	is the length of time of period j
R	is the recovery rate

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